SCATTERPLOTS, CORRELATION & LSRL

new 5.1–5.3, 5.5 • old 2.4–2.6, 2.8

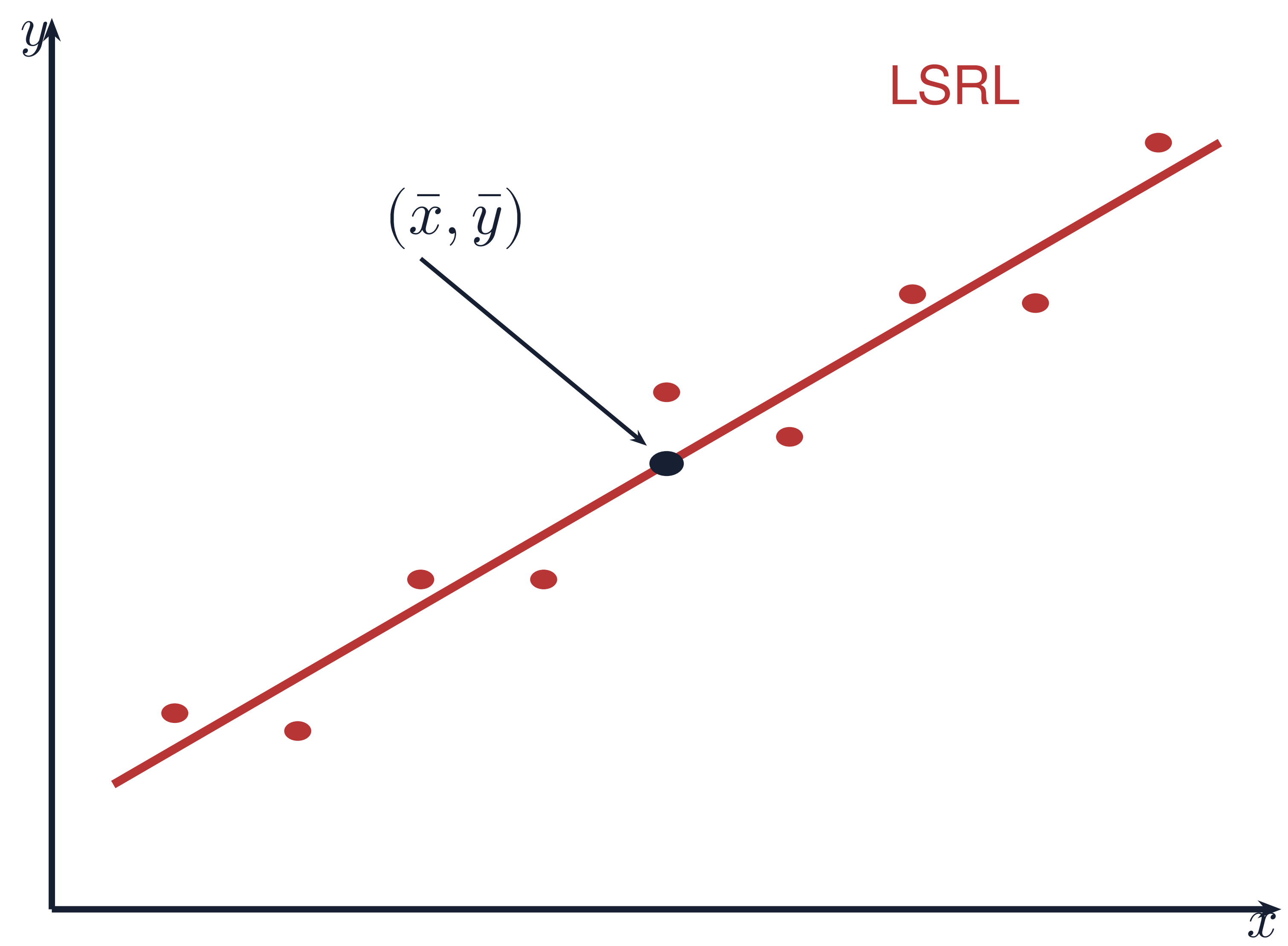
ZONE A · THE LEAST-SQUARES REGRESSION LINE

$$\hat{y} = a + bx$$

$$b = r\frac{s_y}{s_x} \qquad a = \bar{y} - b\bar{x}$$

$$\bar{y} = a + b\bar{x}$$

The LSRL always passes through $(\bar{x}, \bar{y})$. It minimizes the sum of squared vertical residuals.



$$r = \frac{1}{n-1}\sum\left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

DIRECTION · FORM · STRENGTH · UNUSUAL POINTS

Describe the association using all four, with the variables and their context.

ZONE B · WHAT THE SYMBOLS MEAN + HOW r BEHAVES

x: explanatory variable
y: observed response
$\hat{y}$: **predicted** response
a: intercept; b: slope
s_x, s_y: sample SDs; n: number of pairs

$-1 \le r \le 1$; **r has no units.**
Swapping x and y leaves r unchanged.
Changing measurement units leaves r unchanged.
r measures **linear** direction and strength only.
Outliers can change r substantially.

ZONE C · SAY IT IN WORDS

SLOPE

"For each additional ________ [unit of x], the predicted __________ [y] [increases/decreases] by _____ [$|b|$] _______ [units of y]."

INTERCEPT

"When __________ [x] is 0, the predicted __________ [y] is _____ [a] _______ [units]."

This may not make sense in context.

ZONE D · WATCH OUT

Correlation does not establish causation.
Inspect the scatterplot and residual plot (P15) for curvature, even when $|r|$ is large.
Slope has units of y per unit of x; r has no units.

see WS old u2_l4–6 · u2_l8